

LEX PROGRAMS

// Program #1 - Recognizing Numeric Constant: integer / float

```
%{
#include<stdio.h>
%}

%%
[0-9]+      {printf("\n %s is an Integer Number",yytext);}
[0-9]+\.[0-9]+ {printf("\n %s is Float number",yytext);}
.          {}
%%
```

```
void main(){
yylex();
}
```

```
int yywrap() {return 1;}
```

// Program #2 - Recognizing an Identifier

```
%{
#include<stdio.h>
%}
LETTER [a-z]
DIGIT [0-9]

%%
{LETTER}({LETTER}|{DIGIT})* {printf("\n %s is an Identifier",yytext);}
%%
```

```
void main(){
yylex();
}
```

```
int yywrap() {return 1;}
```

//Program #3 - Recognizing Keywords

```
%{
#include<stdio.h>
%}

%%
int |
float |
main {printf("\n %s is a Keyword",yytext);}
\< |
\<= |
\>   {printf("\n %s is a relational operator",yytext);}
.    {printf("%s is not defined",yytext);}
%%
```

```
void main(){
yylex();}
```

```
int yywrap() {return 1;}
```

// Program #4 - Recognizing a Number, from the input file.

```
%{
#include<stdio.h>
%}
%%
[0-9]+      {printf("\n %s is a number",yytext);}
[0-9]+.[0-9]+ {printf("\n %s is a fractional number",yytext);}
.          {printf("\n");}
%%
int main(int argc, char *argv[]){
if(argc>1)
{
yyin=fopen(argv[1],"r");
yylex();
}
return 0;
}
int yywrap() {return 1;}
```

//Program #5 - Counting No of Vowels and Consonents from the given input

```
%{
#include<stdio.h>
int v=0;
int c=0;
%}

%%
[aeiouAEIOU]      {++v;}
.                  {++c;}
"\n"              {printf("\n Vowels:%d\nConsonents:%d",v,c);}
%%
void main(){
yylex();
}

int yywrap() {return 1;}
```

//Program #6 - Counting no of char, words, lines, spaces

```
%{
#include<stdio.h>
int l=0,w=0,c=0,s=0,t=0;
%}
%%
\n      {l++;}
([ ])+  {s++;}
\t      {t++;}
.*      {c++;}
([a-zA-Z0-9])* {w++;}
%%
void main(int argc,char *argv[]){
yyin=fopen(argv[1],"r");
yylex();
```

```

printf("\n Spaces:%d",s);
printf("\n Tabs:%d",t);
printf("\n characters:%d",c);
printf("\n Words:%d",w);
printf("\n Lines:%d",l);
fclose(yyin);
}
int yywrap() {return 1;}

```

//Program #7 - Recognizing Positive or Negative number

```

%{
#include<stdio.h>
%}

%%
[-][0-9]+ {printf("\n %s is a negative integer",yytext);}
[0-9]+    {printf("\n %s is a positive integer",yytext);}
%%
void main()
{
yylex();
}

int yywrap() {return 1;}

```

// Program #8 - Recognizing a valid Mobile number

```

%{
#include<stdio.h>
%}
%%
[1-9][0-9]{9} {printf("\n %s is a valid mobile number",yytext);}
.* {printf("\n Invalid Mobile number");}
%%
void main() {
yylex();
}
int yywrap() {return 1;}

```

//Program #9 - Counting No of lines in the given text file.

```

%{
#include<stdio.h>
int l=1;
%}
line .*\\n
%%
{line} {printf("\n%5d %s",l++,yytext);}
%%
void main(int argc,char *argv[]){
yyin=fopen(argv[1],"r");
yylex();
fclose(yyin);
}
int yywrap() {return 1;}

```

//program #10 - DESIGN OF COMPLETE LEXICAL ANALYZER

```
%{
#include<stdio.h>
%}
letter [a-z]
digit [0-9]

%%
\#.* {printf("\n %s is a Preprocessor Directive",yytext);}
int |
main |
void |
float {printf("\n %s is a Keyword",yytext);}
\( {printf("\n %s is a function",yytext);}
\) ECHO;
\{ {printf("\n %s Block begins",yytext);}
\} {printf("\n %s Block Ends",yytext);}
[0-9]+ {printf("\n %s is a Literal",yytext);}
[0-9]+\.[0-9]+ {printf("\n %s is a real constant",yytext);}
{letter}({letter}|{digit})* {printf("\n %s is an Identifier",yytext);}
= {printf("\n %s is an assignment operator",yytext);}
\+ |
\- |
\* {printf("\n %s is an arithmetic operator",yytext);}
%%
void main(int argc,char *argv[]){
yyin=fopen(argv[1],"r");
yylex();
}
int yywrap() {return 1;}
```

//program #11 - Recognizing and ignoring the comment lines in the program

```
%{
#include<stdio.h>
%}
beginm \/\*
endm \*/
%%
\/\/(.*) {printf("\n Single line comment is ignored);}
{beginm}((.*)"\n"(.*)?)+{endm} {printf("\n Multi line comment is
ignored");}
%%
int main(int argc, char *argv[]){
yyin=fopen(argv[1],"r");
yylex();
fclose(yyin);
return 0;
}

int yywrap() {return 1;}
```

//Program #12 - Printing Line number into the file

```
%{
#include<stdio.h>
int lno=1;
%}
line .*\n
%%
{line}      {printf("\n %5d %s",lno++,yytext);}
%%
int main(int argc,char *argv[]){
yyin=fopen(argv[1],"r");
yylex();
fclose(yyin);
return 0;
}
```

```
int yywrap() {return 1;}
```

// Program #13 - Recognizing an e-mail ID

```
%{
#include<stdio.h>
%}
%%
[a-z][a-z]*(\.)?[a-z0-9]+(@)[a-z]+(\.)([a-z]+|(com|edu)) {printf("\n %s
is a valid ID",yytext);}
. {printf("\n Not a valid ID");}
"\n" {return 0;}
%%
int main(){
yylex();
}
```

```
int yywrap() {return 1;}
```

//Program #14 - Counting no of words

```
%{
#include<stdio.h>
int count=0;
%}

%%
([a-zA-Z0-9])*      {++count;}
"\n" {printf("words are:%d",count);return 0;}
%%
int main(){
yylex();
}
```

```
int yywrap() {return 1;}
```